

ICT361 Handoff — Mulungushi Registrar

Android app

Mulungushi University, School of Engineering and Technology. ICT361 Mobile Application

Development — Android student registration and lab group management (Java, XML Views).

Ported from a web prototype: `campus_roster_combined-48.html` (the version with the plain splash) / `campus_roster_combined-48-1.html` (same app, animated splash — splash itself was not ported, see "Not built" below).

Project name: **MulungushiRegistrar**. Package: `zm.ac.mulungushi.registrar`. Min SDK 24.

Device tested on: TECNO CK6 and a Pixel 5 emulator.

Current blocker — read this first

The last zip (`ict361_home_roster_screens.zip`) fails

`:app:mergeDebugResources` with:

```
Can not extract resource from
com.android.aaptcompiler.ParsedResource@<hash>.,
Can not extract resource from
com.android.aaptcompiler.ParsedResource@<hash>
```

Two instances of this error, different hashes each build. **The message never names the actual file or resource** — that's a known shortcoming of this AGP-internal error, not something either of us is failing to read correctly.

Already ruled out (checked exhaustively, found clean):

- Duplicate `<string>`, `<color>`, `<style>` names across every `values/*.xml` file (this was the cause of an earlier, different, more helpful error — "Found item X more than

one time" — which got fixed; this ParsedResource error is a separate, later problem)

- Duplicate `<item>` entries within a single `<style>` block
- Duplicate drawable/layout filenames across resource folders
- Invalid characters in resource names
- Malformed `layer-list` / `selector` / `shape` XML (checked `bg_radio_on.xml`, `bg_stat_tile.xml`, `color/input_stroke.xml` specifically — all valid syntax)
- UTF-8 BOM on any file
- Bad escaping — `Names can\'t contain numbers.` etc. use the correct single-backslash
Android escape; looked corrupted in one debug pass but was a false alarm from how Python's `repr()` was displaying it, not real corruption
- `night/themes.xml` — nearly empty (just the default wizard comment), not the cause
- **Invalidate Caches / Restart** — tried, did not fix it (this exact fix *did* work earlier in the project for a different error, `cardCornerRadius` not resolving, so it was a reasonable thing to try here too, just didn't apply this time)

Not yet tried — do this next: open the Terminal tab in Android Studio and run

```
gradlew assembleDebug --stacktrace
```

and read the `Caused by:` section, which should contain an actual file path. The Android Studio Build Output panel is truncating/hiding it. The user was mid-way through trying this when the handoff was requested — their next message may already have that output.

If the stack trace still doesn't name a file, the fallback plan (not yet attempted) is bisection: temporarily remove the newest batch of files (everything listed under "Round 3" below) from `res/layout` and the matching Java files, confirm the project builds clean without them, then reintroduce files one at a time to isolate which one triggers it.

What's built, by round

Round 1 — Login. LoginActivity + LoginViewModel + LoginValidator . Demo accounts in

DemoAccounts.java . Fixed early bugs: a themes.xml dotted-style-name linking bug

(Registrar.Label etc. needed explicit parent=""), and later converted away from

MaterialCardView / FloatingActionButton / CoordinatorLayout everywhere in the project after

those repeatedly failed to resolve their app: attributes on this environment even with the

dependency correctly declared — current layouts use plain

FrameLayout / LinearLayout +

background drawables + android:elevation instead, no Material-specific attrs anywhere

except the login screen's TextInputLayout / MaterialButton , which always worked fine.

Round 2 — Student home + Lecturer roster. StudentHomeActivity ,

LecturerRosterActivity ,

StudentAdapter , EditStudentSheet (add/edit/delete), StudentRepository (in-memory,

seeded with 6 students), StudentValidator + its unit test. Also added

ForgotPasswordActivity ,

RegisterActivity , MyGroupActivity somewhere around here (built but the README's manifest

list briefly fell out of sync with reality — now corrected).

Round 3 — everything else (most recent, least field-tested).

GroupChangeActivity ,

EditDetailsActivity , SyncActivity , LecturerDashboardActivity (now the lecturer's actual

sign-in landing screen, replacing a direct-to-roster route), RequestsActivity .

Wired real

navigation through StudentHomeActivity 's cards/bottom-nav and

LecturerRosterActivity 's

bottom-nav, which previously just showed "Not built yet" toasts. **This round has not been**

confirmed working on a device yet — the ParsedResource error above showed up right after

delivering it, so treat everything in this round as unverified until the build is green.

Student model grew two fields this round: pendingGroup , pendingNumber (both String ,

null = no open request). `StudentRepository` grew matching `requestGroupChange / cancelGroupChange / approveGroupChange / declineGroupChange` and the same four for number correction, plus `getPendingRequests()` and `pendingRequestCount()`.

Demo student 202301045 (Chanda Mwansa) is seeded with `pendingGroup = "G02"` so the Pending UI is visible immediately without the user having to create a request first.

Full file list (current)

Java (`app/src/main/java/zm/ac/mulungushi/registrar/`):
`DemoAccounts`, `EditDetailsActivity`, `EditStudentSheet`,
`ForgotPasswordActivity`,
`GroupChangeActivity`, `LecturerDashboardActivity`, `LecturerRosterActivity`,
`LoginActivity`,
`LoginValidator`, `LoginViewModel`, `MyGroupActivity`, `RegisterActivity`,
`RequestsActivity`,
`RosterFormat`, `Student`, `StudentAdapter`, `StudentHomeActivity`,
`StudentRepository`,
`StudentValidator`, `SyncActivity`.

Layouts: `activity_edit_details`, `activity_forgot_password`,
`activity_group_change`,
`activity_lecturer_dashboard`, `activity_lecturer_roster`, `activity_login`,
`activity_my_group`, `activity_register`, `activity_requests`,
`activity_student_home`,
`activity_sync`, `item_student`, `sheet_edit_student`, `top_bar_back` (shared include — crest-free back-arrow top bar used by every non-tab screen).

Tests: `LoginValidatorTest`, `StudentValidatorTest`.

What's simplified / not built (tell the user if asked, don't just silently skip)

- No splash screen (the animated-splash version of the prototype was never ported — not functionally important, purely decorative).
- Requests screen has no "Answered" history tab — only tracks each student's *current*

pending request; once approved/declined there's nothing to look back at.

- Lecturer dashboard's "Conflicts" stat is hardcoded to 0 — real conflict detection needs the offline edit queue from the brief's Activity E (Room transactions, operation IDs, base versions), which doesn't exist yet.
- Lecturer's Sync status screen shows real pending requests, not the prototype's two fixed demo rows (deliberately — showing fake data as if real would be worse than simplifying).
- `StudentRepository` is a plain in-memory `ArrayList`. No Room, no Retrofit, no Node.js/MySQL backend. Resets to the 6 seed students every cold start. This is intentional for now — see the original assignment brief's later activities (B, C, E) for what replaces it.
- `StudentValidator`'s name check keeps the brief's actual required rules (length 2–100, character set, two words) but skips the prototype's cosmetic typing-mistake heuristics (junk-word list, keyboard-mash detection) — see the class's own comment.
- Number-correction request dialog has a number field only, no optional "reason" field the prototype has.
- Tab navigation (Home/Roster/Sync, Home/My group/Sync) doesn't dedupe the back stack — each tap pushes a new activity instance rather than reusing one. Minor polish item, not wired.

Known-good reference points

These were confirmed working on the user's actual device at some point in this build:

- Login screen, including the eye-toggle password field and demo-account hints
- Student home and Lecturer roster screens (Round 2) — confirmed working after several rounds of fixing `MaterialCardView` / `FAB` / `CoordinatorLayout` attribute-resolution failures
- Add/Edit/Delete student sheet, with validation and duplicate-number / full-group checks

- The pending-request indigo banner and pill on Student home (matches the prototype screenshots the user provided)

Setup steps (unchanged across rounds — see README.md in the zip for full detail)

1. Empty Views Activity, Java, package `zm.ac.mulungushi.registrar`, min SDK 24.
2. Drag the zip's `app/src` over the project's `app/src`, replace files.
3. Delete `MainActivity.java` / `activity_main.xml`.
4. Manifest: `ACCESS_NETWORK_STATE` permission, `Theme.Registrar` on `<application>`, `.LoginActivity` as the launcher, then every other activity with `android:exported="false"` and no intent-filter (full list in README.md).
5. Add `androidx.recyclerview:recyclerview` dependency (RecyclerView is used by the roster).
6. Clean Project, Rebuild Project.

Suggested immediate next step for whoever picks this up

1. Get the real `Caused by:` file path via `gradlew assembleDebug --stacktrace` (see "Current blocker" above) before touching any code.
2. Once the build is green, have the user actually exercise Round 3's screens on their device — none of it has been confirmed working yet, only statically checked (XML validity, resource-reference cross-checking, Java brace balance, undefined-method-call scan — all passed, but none of that proves runtime correctness).
3. If the user asks for anything beyond what's listed in "What's simplified" above, check `campus_roster_combined-48.html` directly for the real markup/copy before inventing behavior — this has been the standing rule throughout the project, per the

original
course handoff.